

Refugee Inflows and Native Parity Progression

Evidence from Turkey

Murat Güray Kırdar[†]

M. İkbâl Oruç[†]

[†] Department of Economics, Koç University.

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1. Impact of immigration on native fertility is studied, but **not in the context of refugees**.
2. Fertility rates in OECD countries have fallen by half over the past 60 years to roughly **1.5 children per woman**.
3. **Turkey: a unique case study**
 - Host to one of the world's **largest refugee populations** (≈ 5.3 M Syrians registered, ≈ 3.6 M in Turkey).
 - Native total fertility now **below the OECD average** (≈ 1.41).
 - Ideal setting for studying how migration shocks interact with **low fertility**.

Main data source: 2013 & 2018 rounds of the TDHS.

- Nationally representative microdata on women with rich fertility histories.
 - Year of each birth, age at first cohabitation/marriage, migration history.
- **Pseudo-panel structure:** reconstructed annual marriage and birth histories for women from age 12 to survey date.

Main analytical sample: 2013/18 with **full controls and migration history** for both ever- and never-married women; province assigned by 2011 residence (pre-shock).

Four-wave (2003/08/13/18) ever-married sample used as a complementary specification.

Why FS sample, not EM?

Reconstructing the Pseudo-Panel

Cross-sectional record (TDHS interview):

ID	Birth Yr	Childhood Prov	Move Yr	Dest. Prov	Birth 1	Birth 2
10001	1990	Adana	2009	Mardin	2010	2013

⇓ expanded into woman-by-year panel ⇓

Pseudo-panel:

ID	Age	Year	Province	Gave Birth 1	Gave Birth 2
10001	12	2002	Adana	0	0
⋮	⋮	⋮	⋮	⋮	⋮
10001	20	2010	Mardin	1	0
⋮	⋮	⋮	⋮	⋮	⋮
10001	23	2013	Istanbul	1	1
⋮	⋮	⋮	⋮	⋮	⋮
10001	28	2018	Istanbul	1	1

Cumulative parity outcomes (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \quad p \in \{1, 2, 3, 4, 5\}, \quad a \in \{16, 20, 24, 28, 32, 36\}.$$

Plus marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

Exposure: average refugee-to-native ratio over the woman's age window from 12 to a :

$$\bar{R}_{ja} = \frac{1}{a - 11} \sum_{s=12}^a R_{j, t(s,i)}.$$

Interpretation. A 1 pp. higher *average annual* ratio over ages 12– a raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

Cohort variation. A 2013 woman observed at age 28 averages exposure over 1997–2013 (2 post-shock years); a 2018 woman observed at age 28 averages over 2002–2018 (7 post-shock years). Same age, same province, different $\bar{R}_{ja}$. Crossed with B_j , this cross-cohort variation in window timing is what the IV uses on top of FE.

- **Challenge: endogenous refugee location choice.**
 - Refugees may move toward provinces with better labor-market outcomes.
 - OLS may confound refugee exposure with local economic conditions that also affect fertility.
- **Quasi-exogenous variation.**
 - Early inflows were shaped by **border entry** and **government-run camps near Syria** (2011–2013).
 - This created a persistent **distance gradient**: provinces closer to the Syrian border received higher refugee exposure.
 - As refugees dispersed nationally over time, exposure varied across provinces in a way strongly predicted by **geographic proximity**.

Province-year instrument (Aksu, Erzan and Kirdar 2022):

$$Z_{n,t} = \sum_{s=1}^{13} \underbrace{\left[\frac{(1/d_{s,T}) \pi_s}{\sum_{X \in \{T,L,J,I\}} (1/d_{s,X})} \right]}_{\text{share weight (time-invariant)}} \cdot \frac{T_t}{d_{n,s}}$$

π_s pre-war pop. share of Syrian province s ; $d_{s,X}$ distance from s to host $X \in \{T, L, J, I\}$; $d_{n,s}$ distance from s to Turkish province n ; T_t aggregate Syrian refugee stock across the four host countries.

Key decomposition — only T_t varies in time. Averaging over the woman's exposure window from age 12 to a gives the instrument we use in 2SLS:

$$\bar{Z}_{ja} = B_j \times \bar{T}_a.$$

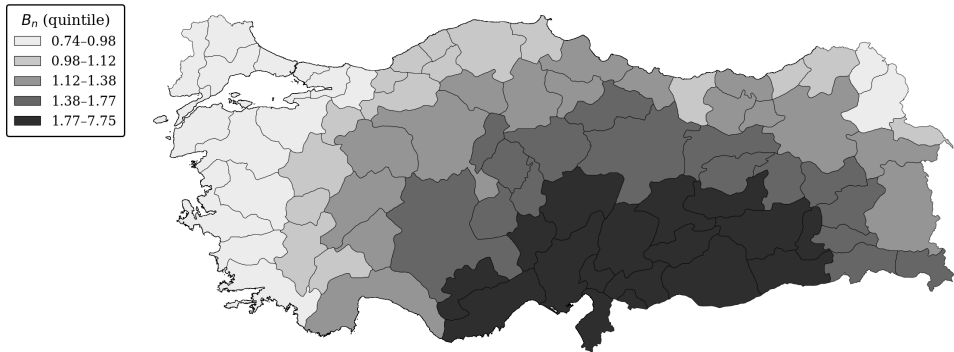
B_j assigns province intensity (geography); $\bar{T}_a$ is cohort-specific via the averaging window.

T_t over time

First-stage F

Settlement-Propensity Weight B_n

Settlement-propensity weight B_n across Turkish provinces (quintiles)



Estimating Equation

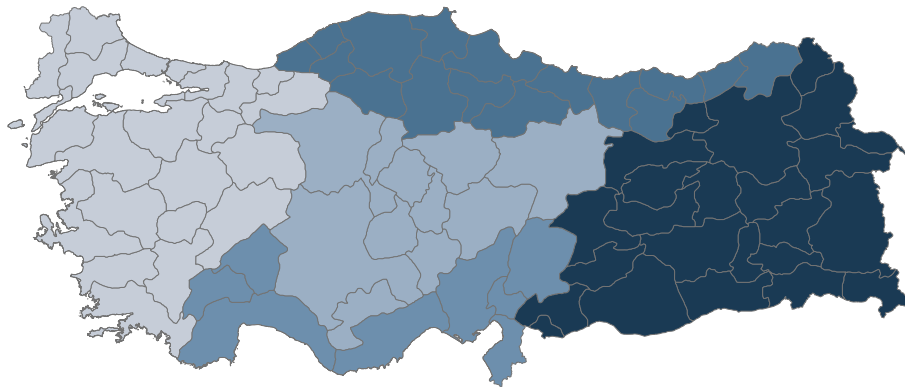
For each (p, a) separately,

$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + X_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a, i) + \lambda_{l(i)} \cdot t(a, i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

- $\beta^{(p,a)}$ — coefficient of interest: effect of a 1 pp. higher *average annual* refugee-to-native ratio on $\Pr(\text{parity} \geq p \text{ by age } a)$.
- X_i' — individual controls: survey wave, childhood urban/rural, mother's education, mother tongue.
- δ_j — province FE; $\mu_{t(a,i)}$ — year-at-age- a FE ($\equiv$ birth-cohort FE at fixed a).
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear time trends.
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear time trends (Turkish / Kurdish / Arabic).
- Trends start in 1987; SEs clustered at province; DHS sampling weights.

Why Region-Specific Trends? — The 5-Region Scheme

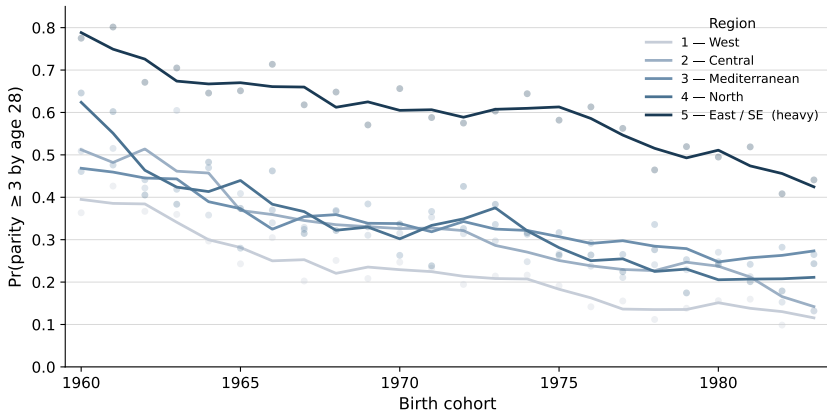


Region

1 — West 2 — Central 3 — Mediterranean 4 — North 5 — East / SE (heavily treated)

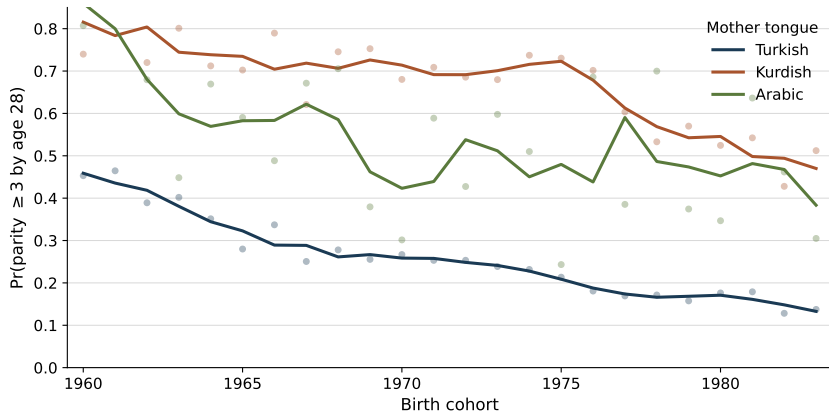
Region 5 hosts $\approx 95\%$ of Syrian refugees in Turkey; regions 1-4 share the remaining $\approx 5\%$.

Why Region-Specific Trends? — Pre-Shock Fertility by Region



Pre-shock cohorts only. **Levels and slopes differ sharply by region:** Region 5 (heavily-treated SE) sits well above the rest throughout and declines more slowly. A common time trend would underfit this differential transition $\Rightarrow$ region-specific trends are needed.

Why Mother-Tongue-Specific Trends?



Pre-shock cohorts only. **Levels and trends differ sharply** across MT groups — and Region 5 (high B_n) is **55% Kurdish**-speaking versus 1.7–12% in regions 1–4. So B_n correlates with MT composition by geography, and MT-specific trends are needed to absorb the differential transition.

Placebo: $Y \sim B_n \times t$ on Pre-Shock Cohorts (BH-FDR)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003 (0.0005)	0.0010 (0.0007)	0.0012 (0.0007)	0.0008 (0.0004)	0.0011 (0.0006)	-0.0001 (0.0007)
Parity ≥ 1	0.0005 (0.0003)	0.0011 (0.0008)	0.0007 (0.0010)	0.0011 (0.0005)	0.0011 (0.0005)	0.0001 (0.0007)
Parity ≥ 2	0.0001 (0.0002)	-0.0001 (0.0004)	0.0009 (0.0006)	0.0018 (0.0007)	0.0015 (0.0010)	0.0006 (0.0012)
Parity ≥ 3	0.0000 (0.0000)	-0.0000 (0.0003)	0.0004 (0.0007)	0.0000 (0.0009)	-0.0007 (0.0008)	-0.0024 (0.0014)
Parity ≥ 4	– (–)	– (–)	0.0002 (0.0004)	0.0011 (0.0009)	0.0005 (0.0023)	-0.0005 (0.0022)
Parity ≥ 5	– (–)	– (–)	0.0000 (0.0002)	0.0002 (0.0005)	0.0012 (0.0014)	0.0018 (0.0023)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Test. On the subsample $\{i : t(a, i) < 2012\}$, regress each outcome on $B_n \times t(a, i)$ with the full set of FE and trends. After BH-FDR adjustment across the 32-cell family (What is BH?), **no cell rejects**. First-stage $F \in [43, 144]$ (FS); four-wave EM appendix.

Main Results — Primary (Full) Sample, BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467 (0.2165)	-0.5715 (0.6000)	0.5130 (1.6892)	-0.0130 (1.0473)	0.2202 (0.9200)	-0.2993 (1.2412)
Parity ≥ 1	0.0095 (0.1273)	0.1226 (0.5231)	-0.0431 (1.9677)	0.9235 (1.3419)	-1.5150 (1.7893)	0.1839 (1.3711)
Parity ≥ 2	0.0285 (0.0409)	0.3048 (0.5464)	-0.1508 (0.7457)	0.6126 (1.9059)	1.7403 (3.4194)	-0.8450 (1.1891)
Parity ≥ 3	0.0084 (0.0055)	0.3816 (0.2481)	-0.3712 (0.5649)	1.2502 (1.5588)	2.8929 (2.6999)	1.0620 (1.8097)
Parity ≥ 4	— (—)	— (—)	0.2496 (0.3399)	1.9047 (0.8094)	2.2458 (1.4146)	4.2520 (1.9632)
Parity ≥ 5	— (—)	— (—)	0.3133 (0.2324)	1.3921 (0.8025)	1.4136 (0.9119)	4.1326 (1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Pooled point estimates positive at higher parities, late 20s/30s; no cell survives BH-FDR across the 32-test family.

Baseline rates

Four-wave (EM)

Heterogeneity I — Heavily-Treated Region (Region 5), BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2452 (0.3582)	-1.6915* (0.6464)	-2.5905 (1.9762)	-0.9127 (1.6556)	-0.4301 (0.7606)	1.5648 (0.8548)
Parity ≥ 1	-0.0233 (0.1541)	-0.4039 (0.4780)	-3.0301 (1.5690)	1.4459 (2.2608)	0.3391 (1.4282)	1.9888 (1.1851)
Parity ≥ 2	0.0210 (0.0518)	1.4250** (0.4706)	1.2829 (0.9335)	0.4696 (2.1259)	5.4953* (2.4693)	0.3303 (1.1054)
Parity ≥ 3	-0.0151 (0.0193)	0.3390 (0.4285)	-0.0495 (0.5988)	5.1117* (2.0445)	9.1213** (2.8160)	4.4002* (1.7627)
Parity ≥ 4	— (—)	— (—)	0.2292 (0.5746)	2.9874* (1.2826)	6.7512*** (1.6983)	5.7527* (2.4905)
Parity ≥ 5	— (—)	— (—)	-0.0136 (0.4266)	1.8753 (0.9337)	4.1483** (1.3901)	6.5940 (3.4988)
Observations	4,002	3,522	3,022	2,404	1,851	1,311

Effects **concentrate in the heavily-treated southeast**, at higher parities and late 20s/30s. Other regions essentially null (**regions 1–4**). **Baseline rates**

Heterogeneity II — Kurdish / Arabic Mother Tongue, BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.5562 (0.3233)	0.5199 (1.7594)	0.5854 (2.4777)	-4.0251 (2.2391)	1.9332 (2.2384)	0.8593 (1.1457)
Parity ≥ 1	0.3356 (0.1769)	1.8205 (1.5930)	-1.7541 (2.3234)	-1.5852 (2.3963)	2.4411 (2.7699)	0.7623 (1.8112)
Parity ≥ 2	0.0471 (0.0729)	1.8055 (1.1665)	1.4326 (1.8375)	1.4601 (2.9870)	4.4175 (3.2879)	1.9937 (1.4012)
Parity ≥ 3	-0.0192 (0.0258)	0.6474 (1.2286)	-0.8687 (1.2296)	12.3048*** (2.4693)	6.0191 (3.4792)	2.5694 (1.5223)
Parity ≥ 4	— (—)	— (—)	1.2013 (1.3175)	8.1188*** (1.7515)	2.0236 (3.8106)	8.2440*** (1.7884)
Parity ≥ 5	— (—)	— (—)	0.3944 (1.0175)	3.1548 (2.4496)	4.7186 (3.3471)	5.0376 (2.3657)
Observations	3,145	2,785	2,415	1,951	1,489	1,055

Sharp signal among Kurdish/Arabic women at higher parities and late 20s/30s. Turkish-speaking women effectively null (Turkish). Baseline rates

Heterogeneity III — Lower-Educated Women ($\leq$ middle school), BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.1848 (0.3031)	-0.8722 (1.2854)	2.0480 (1.8239)	-0.0590 (1.5294)	0.1893 (1.0346)	-0.2025 (1.1901)
Parity ≥ 1	-0.0234 (0.1870)	0.0484 (1.2489)	0.2907 (2.4427)	1.3034 (1.9572)	0.0446 (1.2737)	0.5583 (1.3313)
Parity ≥ 2	0.0029 (0.0571)	0.2609 (1.6357)	-1.1164 (1.6824)	1.6909 (1.6453)	1.6018 (2.7430)	1.9856 (1.6354)
Parity ≥ 3	0.0038 (0.0069)	0.7375 (0.7647)	-0.6375 (1.2099)	2.8227 (2.2384)	3.2607 (2.4464)	4.6894** (1.6967)
Parity ≥ 4	— (—)	— (—)	-0.0667 (0.6786)	3.1747* (1.2937)	2.9678** (1.0527)	6.6579** (2.3695)
Parity ≥ 5	— (—)	— (—)	0.3507 (0.4137)	1.7869 (1.2837)	1.8934 (1.1604)	5.6194** (1.8604)
Observations	9,361	9,132	8,729	7,633	6,348	4,897

Effect concentrated in **lower-educated** women at higher parities and late 20s/30s. High-education women null (**high educ**). **Baseline rates** **Common thread across heterogeneity:** effects appear in **higher-baseline-fertility** groups where there is room to move up the parity ladder.

- **Labor-market / opportunity-cost mechanism** (Aksu, Erzan and Kirdar 2022):
 - Refugee inflows displace native women's jobs disproportionately — informal-sector substitution.
 - Lower native female employment $\Rightarrow$ **lower opportunity cost of childbearing** relative to work.
 - Predicts a shift in marriage timing and parity progression for women whose labor-supply margin is most affected.
- **Social-norm / preference-shift mechanism:**
 - Large demographic shocks can change perceived returns to marriage and fertility, or shift local norms about family size.
 - Potentially strongest in regions with high refugee exposure and among groups with **early-marriage and high-fertility norms**.
 - Testable implication: a shift in *stated* fertility preferences (ideal n) at the same parities where realised fertility responds.

Social-Norm Mechanism: Stated Fertility Preferences (Ideal n)

Outcome. $\mathbb{1}\{n_i^* \geq k\}$, where n_i^* is woman i 's stated ideal number of children, for $k \in \{2, 3, 4, 5\}$. Cross-sectional 2SLS, four-wave EM sample. For each k :

$$\mathbb{1}\{n_i^* \geq k\} = \alpha + \beta_k R_{j(i), t(i)} + X_i' \Gamma + \delta_j + \mu_t + \theta_{g(j)} t + \lambda_{l(i)} t + \varepsilon_i.$$

Subgroup	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
Region 5 (heavy)	0.0688 (0.1169)	0.2123 (0.2372)	0.7273** (0.3361)	0.3676** (0.1428)
Kurdish / Arabic	0.1287 (0.1578)	0.5124 (0.4172)	0.9109*** (0.3501)	0.4754*** (0.1844)
Observations	4,003 (Region 5); 3,177 (Kurdish / Arabic)			

Stated preferences shift up at the upper thresholds in the same subgroups where realised fertility responds. A shift in *desired* family size is hard to rationalise as a pure labor-market response (which would change *behavior*, not preferences); more naturally read as a **norm / preference shift**. Pooled and other subgroups in [appendix](#).

The Syrian inflow raised native parity progression in groups with higher baseline fertility.

- Pooled effects positive at higher parities, late 20s/30s; no cell survives BH-FDR across 32 tests.
- Under BH-FDR, the response is sharpest in three subgroups: **heavily-treated southeast, Kurdish/Arabic** mother-tongue women, and **lower-educated** women — all groups with higher baseline fertility.
- Stated **ideal- n** shifts up at the same upper thresholds in the same subgroups.
- Marriage and first births unaffected everywhere.

Mechanism. Pattern is consistent with both a **labor-market / opportunity-cost** channel (refugee inflows reduce female employment in the affected groups, lowering the cost of additional children) and a **social-norm / preference-shift** channel (the shift in stated ideal- n at the same parities). Disentangling the two is ongoing work.

Thank you. / Questions?

Appendix

Appendix: Why FS Sample (2013/18), not Four-Wave EM?

The four-wave EM sample adds 2003 and 2008 (pre-shock) cohorts but pays two prices:

- **Sample selection.** EM conditions on ever-married. Marriage is itself an outcome that may respond to the treatment, so the EM sample composition can shift with treatment.
- **Province measurement error.** 2003/08 waves lack detailed migration histories, so EM must use *childhood province* (province at age 12). A woman who moved as an adult is misclassified — her fertility decisions happened at her adult province, not her childhood one. Classical measurement error $\Rightarrow$ **attenuation bias toward zero**.
- **Limited controls.** 2003/08 only have detailed individual controls (mother's education, mother tongue, childhood urban/rural) for ever-married women.

FS sample (2013/18 only): both ever- and never-married women; province by 2011 residence (predetermined, correct for the exposure years); full controls for everyone.

Trade-off. FS = cleaner causal interpretation. EM = more pre-shock anchoring at the cost of selection + attenuation bias. We lead with FS and report EM as a complementary specification.

Appendix: What is BH-FDR Adjustment?

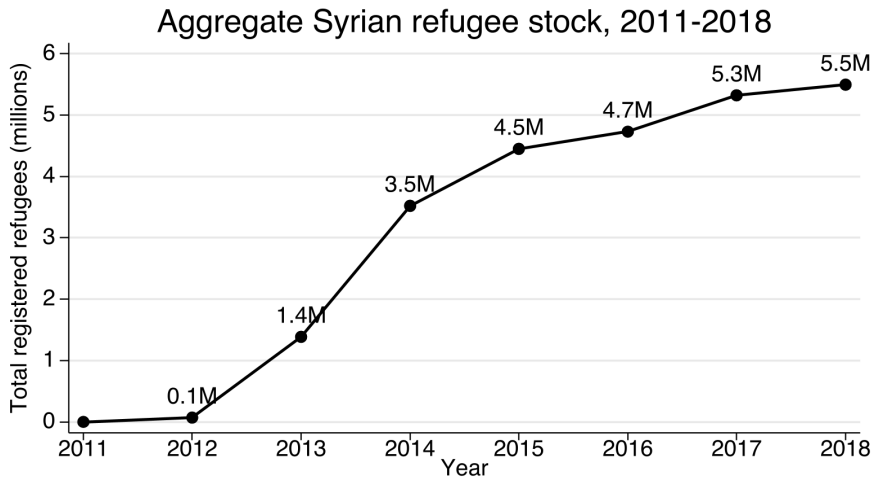
Problem. The pooled fertility analysis runs $5 \times 6 = 30$ tests (parities $\times$ ages); plus 2 marriage cells = 32. At $\alpha = 0.05$, we expect ~ 1.6 false positives by chance even if every true effect is zero.

Benjamini–Hochberg (BH) controls the **false discovery rate**:

$$\text{FDR} = \mathbb{E} \left[\frac{\# \text{false rejections}}{\# \text{rejections}} \right] \leq \alpha.$$

Of the cells we call significant, on average a fraction α are false.

Recipe. Sort raw p -values $p_{(1)} \leq \dots \leq p_{(m)}$. Reject all hypotheses through the largest k with $p_{(k)} \leq \frac{k}{m} \alpha$. Equivalently, report adjusted $p_{(k)}^{\text{BH}} = \min_{j \geq k} \frac{m}{j} p_{(j)}$.



Primary sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	43.93	131.31	102.39	43.72	52.90	143.70
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Four-wave (EM) sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	29.40	32.91	51.24	43.49	53.01	83.21
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Placebo — Four-Wave (EM), BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.0003 (0.0002)	0.0004 (0.0005)	0.0003 (0.0005)	0.0008 (0.0004)	0.0006* (0.0002)	0.0002 (0.0002)
Parity ≥ 2	0.0001 (0.0001)	0.0002 (0.0003)	0.0004 (0.0003)	0.0003 (0.0005)	0.0014* (0.0005)	0.0005 (0.0003)
Parity ≥ 3	0.0000 (0.0000)	0.0002 (0.0001)	-0.0002 (0.0004)	-0.0009 (0.0004)	-0.0007 (0.0003)	-0.0008 (0.0006)
Parity ≥ 4	— (—)	— (—)	-0.0004 (0.0002)	-0.0011* (0.0004)	-0.0013 (0.0008)	-0.0016 (0.0009)
Parity ≥ 5	— (—)	— (—)	0.0002 (0.0001)	-0.0009** (0.0003)	-0.0013 (0.0008)	-0.0024*** (0.0006)
Observations	18,943	20,519	20,204	17,909	14,367	10,350

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Appendix: Main IV — Four-Wave (EM), BH-FDR

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.3132 (1.9142)	-1.3481 (1.2286)	0.9856 (2.2382)	-0.6077 (1.1088)	-1.3975 (1.4367)	1.3735 (0.6797)
Parity ≥ 2	0.3222 (0.1720)	1.2727 (2.2397)	0.6867 (1.4517)	-1.0745 (2.0183)	2.2526 (2.6152)	0.0998 (1.9274)
Parity ≥ 3	0.1429 (0.1168)	1.4820 (0.9743)	-0.7720 (0.9236)	-0.9003 (1.9910)	0.7142 (2.3653)	-0.5607 (2.5075)
Parity ≥ 4	— (—)	— (—)	-0.3685 (0.5178)	0.4007 (0.7539)	0.0293 (1.5816)	2.7419 (2.0043)
Parity ≥ 5	— (—)	— (—)	0.2797 (0.3048)	0.7892 (0.5506)	-0.3457 (1.2736)	1.5672 (1.6244)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

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Appendix: Region 1–4 (“Other”), Primary (BH-FDR)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2609 (0.1868)	-1.3578 (1.0728)	1.7261 (1.7837)	0.0472 (1.2570)	-2.0559 (1.6061)	-1.9054* (0.7543)
Parity ≥ 1	0.0452 (0.2257)	-0.3939 (1.0197)	1.8234 (1.4001)	1.3042 (2.0493)	-6.0697 (2.5172)	-0.9893 (0.9780)
Parity ≥ 2	-0.0363 (0.0429)	-0.5946 (0.6160)	-1.2016 (0.9670)	2.6652 (2.1099)	-4.8945 (3.1711)	-1.1427 (1.6412)
Parity ≥ 3	-0.0006 (0.0014)	0.1853 (0.3480)	-0.6541 (0.7896)	-0.1352 (3.9169)	0.1920 (3.8745)	0.5938 (2.6857)
Parity ≥ 4	– (–)	– (–)	0.2275 (0.4780)	1.4477 (1.3774)	1.4941 (2.3666)	6.1002 (2.5449)
Parity ≥ 5	– (–)	– (–)	0.5797 (0.2703)	0.8921 (1.3467)	1.4853 (1.6360)	5.4374 (2.7984)
Observations	10,352	9,962	9,219	7,836	6,394	4,799

Appendix: Turkish Mother-Tongue Women, Primary (BH-FDR)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.4330 (0.1970)	-1.2864 (0.7963)	-0.1425 (1.6814)	0.2402 (1.6210)	-1.5190 (1.2953)	-2.3851 (2.9491)
Parity ≥ 1	-0.1718 (0.1202)	-0.8128 (0.7729)	-0.5693 (2.3789)	1.2331 (1.9282)	-3.9264 (2.7780)	-2.1423 (3.3247)
Parity ≥ 2	-0.0034 (0.0239)	0.0618 (0.7466)	-1.2228 (0.9056)	0.1915 (2.7639)	-0.7790 (5.4613)	-3.6633 (2.3983)
Parity ≥ 3	0.0041 (0.0043)	0.1653 (0.3170)	-0.3431 (0.5697)	-1.0986 (1.7848)	2.7682 (4.0056)	-0.8493 (3.3966)
Parity ≥ 4	– (–)	– (–)	-0.0748 (0.3250)	-0.3461 (1.0642)	2.5819 (1.0607)	1.8207 (2.9586)
Parity ≥ 5	– (–)	– (–)	0.2606 (0.1146)	-0.0430 (0.3103)	0.0264 (1.5401)	5.1158 (3.8226)
Observations	11,013	10,512	9,655	8,137	6,634	4,960

Appendix: Higher-Educated Women, Primary (BH-FDR)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0176 (0.0534)	0.2143 (0.5891)	-0.8859 (3.0142)	-1.4122 (3.2686)	-1.6326 (2.9202)	1.9265 (4.8016)
Parity ≥ 1	0.0414 (0.0289)	-0.3907 (0.6715)	-0.5787 (2.9107)	0.4812 (3.7083)	-9.4252 (4.9289)	0.6246 (5.4977)
Parity ≥ 2	0.0006 (0.0235)	0.3035 (0.2388)	0.2737 (1.1549)	0.1005 (4.6549)	-4.5550 (7.5530)	-15.0970 (8.5615)
Parity ≥ 3	- (-)	0.0801 (0.0614)	0.0441 (0.8062)	-3.2287 (1.2656)	7.3840 (6.9828)	-18.7862 (7.4317)
Parity ≥ 4	- (-)	- (-)	-0.5710 (0.4974)	-0.3215 (0.3705)	3.9271 (3.1890)	-8.8020 (4.8629)
Parity ≥ 5	- (-)	- (-)	-0.4857 (0.4814)	0.1304 (0.1855)	-0.3653 (0.5670)	-2.3138 (2.1605)
Observations	4,993	4,352	3,512	2,607	1,897	1,213

Appendix: Ideal Fertility — Pooled (Four-Wave EM)

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0458 (0.0605)	0.1739 (0.2536)	0.5142* (0.2700)	0.2555** (0.1087)
Observations	14,775	14,775	14,775	14,775

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Appendix: Baseline Rates — Overall (Full Sample)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	10.0%	45.2%	73.8%	89.2%	93.6%	95.8%
Ever Gave Birth to One Child	3.3%	31.1%	61.5%	80.0%	88.8%	92.1%
Ever Gave Birth to Two Children	0.4%	9.9%	34.3%	56.8%	71.8%	80.3%
Ever Gave Birth to Three Children	0.0%	1.9%	11.0%	24.2%	36.7%	44.8%
Ever Gave Birth to Four Children	0.0%	0.4%	3.7%	10.3%	15.9%	21.1%
Ever Gave Birth to Five Children	0.0%	0.1%	1.1%	4.5%	8.0%	11.2%
Observations	13,034	12,817	12,007	10,433	8,442	6,295

Cells: share of women in each age column who have reached the row outcome.

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Appendix: Baseline Rates — Heavily-Treated Region 5

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	12.9%	50.6%	75.3%	87.7%	92.9%	95.4%
Ever Gave Birth to One Child	4.6%	36.5%	65.5%	82.0%	89.6%	92.4%
Ever Gave Birth to Two Children	0.9%	17.3%	48.1%	69.6%	81.6%	88.2%
Ever Gave Birth to Three Children	0.1%	4.8%	24.5%	48.8%	64.8%	74.9%
Ever Gave Birth to Four Children	0.0%	1.4%	11.9%	29.7%	44.2%	58.2%
Ever Gave Birth to Five Children	0.0%	0.3%	3.8%	15.9%	28.7%	41.0%
Observations	3,757	3,406	2,981	2,453	1,902	1,350

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Appendix: Baseline Rates — Kurdish / Arabic Mother Tongue

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.6%	55.0%	78.5%	89.6%	93.6%	96.4%
Ever Gave Birth to One Child	5.5%	41.3%	69.7%	84.0%	90.5%	93.6%
Ever Gave Birth to Two Children	1.2%	20.4%	54.0%	74.1%	84.6%	90.9%
Ever Gave Birth to Three Children	0.1%	6.0%	28.8%	53.6%	69.4%	78.7%
Ever Gave Birth to Four Children	0.0%	1.8%	14.1%	33.9%	48.3%	60.6%
Ever Gave Birth to Five Children	0.0%	0.3%	4.7%	18.9%	31.8%	43.8%
Observations	2,940	2,722	2,396	1,987	1,526	1,090

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Appendix: Baseline Rates — Lower-Educated Women

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	16.1%	61.5%	85.5%	93.5%	96.0%	97.4%
Ever Gave Birth to One Child	5.4%	44.7%	76.1%	87.9%	92.6%	94.5%
Ever Gave Birth to Two Children	0.7%	15.4%	47.1%	69.9%	81.4%	86.5%
Ever Gave Birth to Three Children	0.0%	2.9%	16.0%	32.5%	46.7%	53.6%
Ever Gave Birth to Four Children	0.0%	0.7%	5.5%	14.2%	21.2%	26.6%
Ever Gave Birth to Five Children	0.0%	0.1%	1.6%	6.3%	10.8%	14.4%
Observations	8,316	8,514	8,468	7,739	6,477	5,026

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Appendix: Ideal Fertility — Other Subgroups

Lower-educated women:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0012 (0.0879)	0.2032 (0.2468)	0.6757*** (0.2471)	0.3513*** (0.1005)
Observations	11,723	11,723	11,723	11,723

Childhood village:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	-0.1961* (0.1081)	-0.0944 (0.2524)	0.5148 (0.4037)	0.2867 (0.2053)
Observations	7,275	7,275	7,275	7,275